

Section: SC-60

D-BLOCK COMPOUNDS

Date: 26-06-2020

NAME:

ID.NO:

ONE OR MORE THAN ONE CORRECT OPTION TYPE

- (T) imparts violet colour $\xrightarrow{\text{compd (U)} + \text{conc. H}_2\text{SO}_4}$ (V) Red gas $\xrightarrow{\text{NaOH} + \text{AgNO}_3}$ (W) Red ppt. $\xrightarrow{\text{NH}_3 \text{ soln.}}$ (X)

(W) Red ppt. $\xrightarrow{\text{dil. HCl}}$ (Y) white ppt.

(U) $\xrightarrow[\Delta]{\text{NaOH}}$ (Z) gas (gives white fumes with HCl)

sublimes on heating

Identify (T) to (Z).

(A) T = KMnO_4 , U = HCl, V = Cl_2 , W = HgI_2 , X = $\text{Hg}(\text{NH}_2)\text{NO}_3$, Y = Hg_2Cl_2 , Z = N_2

(B) T = $\text{K}_2\text{Cr}_2\text{O}_7$, U = NH_4Cl , V = CrO_2Cl_2 , W = Ag_2CrO_4 , X = $[\text{Ag}(\text{NH}_3)_2]^+$, Y = AgCl , Z = NH_3

(C) T = K_2CrO_4 , U = KCl, V = CrO_2Cl_2 , W = HgI_2 , X = Na_2CrO_4 , Y = BaCO_3 , Z = NH_4Cl

(D) T = K_2MnO_4 , U = NaCl, V = CrO_3 , W = AgNO_2 , X = $(\text{NH}_4)_2\text{CrO}_4$, Y = CaCO_3 , Z = SO_2
- The number of moles of acidified KMnO_4 required to convert one mole of sulphite ion into sulphate ion is

(A) $2/5$ (B) $3/5$ (C) $4/5$ (D) 1
- $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \xrightleftharpoons[\text{Mo}]{\text{Fe}}$ $2\text{NH}_3(\text{g})$; Haber's process, Mo is used as

(A) a catalyst (B) a catalytic promoter

(C) an oxidising agent (D) as a catalytic poison
- Potash alum is a double salt, its aqueous solution shows the characteristics of

(A) Al^{3+} ions (B) K^+ ions (C) SO_4^{2-} ions (D) Al^{3+} ions but not K^+ ions
- $\text{Cr}_2\text{O}_7^{2-} \xrightleftharpoons[\text{Y}]{\text{X}} 2\text{CrO}_4^{2-}$, X and Y are respectively

(A) X = OH^- , Y = H^+ (B) X = H^+ , Y = OH^-

(C) X = OH^- , Y = H_2O_2 (D) X = H_2O_2 , Y = OH^-
- Addition of non-metals like B and C to the interstitial sites of a transition metal results the metal

(A) of more ductability (B) of less ductability (C) less malleable (D) of more hardness
- Mercury is a liquid at 0°C because of

(A) very high ionisation energy (B) weak metallic bonds

(C) high heat of hydration (D) high heat of sublimation
- CrO_3 dissolves in aqueous NaOH to give

(A) $\text{Cr}_2\text{O}_7^{2-}$ (B) CrO_4^{2-} (C) $\text{Cr}(\text{OH})_3$ (D) $\text{Cr}(\text{OH})_2$
- Solution of MnO_4^- is purple-coloured due to

(A) d-d-transition (B) charge transfer from O to Mn

(C) due to both d-d-transition and charge transfer (D) none of these
- During estimation of oxalic acid Vs KMnO_4 , self indicator is

(A) KMnO_4 (B) oxalic acid (C) K_2SO_4 (D) MnSO_4

11. Correct statement(s) is/are
 (A) an acidified solution of $K_2Cr_2O_7$ liberates iodine from KI
 (B) $K_2Cr_2O_7$ is used as a standard solution for estimation of Fe^{2+} ions
 (C) in acidic medium, $M = N/6$ for $K_2Cr_2O_7$
 (D) $(NH_4)_2Cr_2O_7$ on heating decomposes to yield Cr_2O_3 through an endothermic reaction
12. A compound of mercury used in cosmetics, in Ayurvedic and Yunani medicines and known as Vermilion is
 (A) $HgCl_2$ (B) HgS (C) Hg_2Cl_2 (D) HgI
13. Acidified chromic acid + $H_2O_2 \xrightarrow[\text{(blue colour)}]{\text{Org. solvent}} X + Y$, X and Y are
 (A) CrO_5 and H_2O (B) Cr_2O_3 and H_2O (C) CrO_2 and H_2O (D) CrO and H_2O
14. $\uparrow Y(g) \xleftarrow{KI} CuSO_4 \xrightarrow{dil H_2SO_4} X(\text{Blue colour})$, X and Y are
 (A) $X = I_2$, $Y = [Cu(H_2O)_4]^{2+}$ (B) $X = [Cu(H_2O)_4]^{2+}$, $Y = I_2$
 (C) $X = [Cu(H_2O)_4]^+$, $Y = I_2$ (D) $X = [Cu(H_2O)_5]^{2+}$, $Y = I_2$
15. $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is
 (A) Cr_2O_3 (B) CrO_2 (C) Cr_2O_4 (D) CrO_3
16. Iron becomes passive by due to formation of
 (A) dil. HCl , Fe_2O_3 (B) 80% conc. HNO_3 , Fe_3O_4
 (C) conc. H_2SO_4 , Fe_3O_4 (D) conc. HCl , Fe_3O_4
17. Bayer's reagent used to detect olifinic double bond is
 (A) acidified $KMnO_4$ (B) aqueous $KMnO_4$
 (C) 1% alkaline $KMnO_4$ solution (D) $KMnO_4$ in benzene
18. $MnO_4^- + xe^- \xrightarrow{\text{(Alkaline medium)}} MnO_4^{2-}$
 $\quad \quad \quad + ye^- \xrightarrow{\text{(Acidic medium)}} Mn^{2+}$
 $\quad \quad \quad + ze^- \xrightarrow{\text{(Neutral medium)}} MnO_2$
 x, y and z are respectively
 (A) 1, 2, 3 (B) 1, 5, 3 (C) 1, 3, 5 (D) 5, 3, 1
19. $Cu + \text{conc. } HNO_3 \xrightarrow{\text{(hot)}} Cu(NO_3)_2 + X$ (oxide of nitrogen); then X is
 (A) N_2O (B) NO_2 (C) NO (D) N_2O_3
20. When $KMnO_4$ solution is added to hot oxalic acid solution, the decolourisation is slow in the beginning but becomes instantaneous after some time. This is because
 (A) Mn^{2+} acts as auto catalyst (B) CO_2 is formed
 (C) Reaction is exothermic (D) MnO_4^- catalyses the reaction.
21. $CuSO_4$ solution reacts with excess KCN to give
 (A) $Cu(CN)_2$ (B) $CuCN$ (C) $K_2[Cu(CN)_2]$ (D) $K_3[Cu(CN)_4]$
22. Pick out the incorrect statement:
 (A) MnO_2 dissolves in conc. HCl , but does not form Mn^{4+} ions
 (B) MnO_2 oxidizes hot concentrated H_2SO_4 liberating oxygen
 (C) K_2MnO_4 is formed when MnO_2 in fused KOH is oxidised by air, KNO_3 , PbO_2 or $NaBiO_3$
 (D) Decomposition of acidic $KMnO_4$ is not catalysed by sunlight.
23. 1 mole of Fe^{2+} ions are oxidised to Fe^{3+} ions with the help of (in acidic medium)
 (A) 1/5 moles of $KMnO_4$ (B) 5/3 moles of $KMnO_4$
 (C) 2/5 moles of $KMnO_4$ (D) 5/2 moles of $KMnO_4$
24. The metals present in insulin and haemoglobin are respectively
 (A) Zn, Hg (B) Zn, Fe (C) Co, Fe (D) Mg, Fe

25. To an acidified dichromate solution, a pinch of Na_2O_2 is added and shaken. What is observed:
 (A) blue colour (B) Orange colour changing to green
 (C) Copious evolution of oxygen (D) Bluish - green precipitate
26. Amongst CuF_2 , CuCl_2 and CuBr_2
 (A) only CuF_2 is ionic
 (B) both CuCl_2 and CuBr_2 are covalent
 (C) CuF_2 and CuCl_2 are ionic but CuBr_2 is covalent
 (D) CuF_2 , CuCl_2 as well as CuBr_2 are ionic
27. A metal M which is not affected by strong acids like conc. HNO_3 , conc. H_2SO_4 and conc. solution of alkalies like NaOH , KOH forms MCl_3 which finds use for toning in photography. The metal M is
 (A) Ag (B) Hg (C) Au (D) Cu
28. Solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ having covalent, ionic as well as co-ordinate bonds. Copper atom/ion forms co-ordinate bonds with water.
 (A) 1 (B) 2 (C) 3 (D) 4
29. $\text{CuSO}_4(\text{aq}) + 4\text{NH}_3 \longrightarrow \text{X}$, then X is
 (A) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (B) paramagnetic
 (C) coloured (D) of a magnetic moment of 1.73 BM
30. $\text{KMnO}_4 + \text{HCl} \longrightarrow \text{H}_2\text{O} + \text{X}(\text{g})$, X is a (acidified)
 (A) red liquid (B) violet gas (C) greenish yellow gas (D) yellow-brown gas
31. Number of moles of SnCl_2 required for the reduction of 1 mole of $\text{K}_2\text{Cr}_2\text{O}_7$ into Cr_2O_3 is (in acidic medium)
 (A) 3 (B) 2 (C) 1 (D) 1/3
32. Pick out the incorrect statement:
 (A) MnO_4^{2-} is quite strongly oxidizing and stable only in very strong alkalies. In dilute alkali, neutral solutions, it disproportionates.
 (B) In acidic solutions, MnO_4^- is reduced to Mn^{2+} and thus, KMnO_4 is widely used as oxidising agent
 (C) KMnO_4 does not acts as oxidising agent in alkaline medium
 (D) KMnO_4 is manufactured by the fusion of pyrolusite ore with KOH in presence of air or KNO_3 , followed by electrolytic oxidation in alkaline solution.
33. The aqueous solution of CuCrO_4 is green because it contains
 (A) green Cu^{2+} ions (B) green CrO_4^{2-} ions
 (C) blue Cu^{2+} ions and green CrO_4^{2-} ions (D) blue Cu^{2+} ions and yellow CrO_4^{2-} ions
34. Acidified KMnO_4 can be decolourised by
 (A) SO_2 (B) H_2O_2 (C) FeSO_4 (D) $\text{Fe}_2(\text{SO}_4)_3$
35. Colourless solutions of the following four salts are placed separately in four different test tubes and a strip of copper is dipped in each one of these. Which solution will turn blue?
 (A) KNO_3 (B) AgNO_3 (C) $\text{Zn}(\text{NO}_3)_2$ (D) ZnSO_4
36. Iron salt used in blue prints is:
 (A) FeC_2O_4 (B) $\text{Fe}_2(\text{C}_2\text{O}_4)_3$ (C) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (D) $\text{K}_3[\text{Fe}(\text{CN})_6]$
37. When acidified KMnO_4 is added to hot oxalic acid solution, the decolourization is slow in the beginning, but becomes very rapid after some time. This is because:
 (A) Mn^{2+} acts as autocatalyst (B) CO_2 is formed as the product
 (C) Reaction is exothermic (D) MnO_4^- catalyses the reaction
38. **Assertion :** KMnO_4 is purple in colour due to charge transfer.
Reason : In MnO_4^- , there is no electron present in d-orbitals of manganese.
39. **Assertion :** K_2CrO_4 has yellow colour due to charge transfer.

Reason : CrO_4^{2-} ion is tetrahedral in shape.

40. **Assertion :** The highest oxidation state of chromium in its compounds is +6.

Reason : Chromium atom has only six electrons in ns and (n-1) d orbitals.

41. **Assertion :** CrO_3 reacts with HCl to form chromyl chloride gas.

Reason : Chromyl chloride (CrO_2Cl_2) has tetrahedral shape.

42. **Assertion :** Equivalent mass of KMnO_4 is equal to one-third of its molecular mass when it acts as an oxidising agent in an alkaline medium.

Reason : Oxidation number of Mn is +7 in KMnO_4 .

KEY

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|-----|-------|-----|---------|-----|---|-----|-------|-----|-------|-----|-------|
| 1. | B | 2. | A | 3. | B | 4. | A,B,C | 5. | A | 6. | B,C,D |
| 7. | A,B | 8. | B | 9. | B | 10. | A | 11. | A,B,C | 12. | B |
| 13. | B | 14. | A | 15. | B | 16. | C | 17. | B | 18. | A |
| 19. | D | 20. | D | 21. | A | 22. | B | 23. | A,C | 24. | B |
| 25. | D | 26. | A,B,C,D | 27. | C | 28. | A | 29. | C | 30. | D |
| 31. | A,B,C | 32. | B | 33. | B | 34. | A | 35. | B | 36. | B |
| 37. | B | 38. | B | 39. | B | 40. | A | | | | |